

Game 07 — Spirit Runner (Complete Plan)

Game 7 of 7 Complexity: ★★★ High Estimate: 5–7 days Status: Not started

Complete plan (supersedes the 2026-09-09 sample). Status: **not started** — build-ready, **requires Game 05's engine first** (forks `engine.js`). Slug `spirit-runner`; folder `apps/frontend/public/games/spirit-runner/`. Highest complexity (5–7 days); phased A–D so every phase ends shippable.

1. Overview

Mystical-forest endless runner with light puzzle choices: run, dodge, collect spirit orbs for powers, and at path splits pick the right **rune gate** — wrong pick banishes you to the riskier, richer **shadow realm**. Depth from choices, not controls (one thumb).

2. Complete game spec

Phase A — core run (fork of Game 05)

- Jump (as 05: coyote/buffer/cut) **plus slide**: swipe down / ↓ / S — 600 ms, hitbox drops to 40 % height; slide-jump cancels into a hop.
- Obstacles:

Type	Clearance	Look (procedural)	Introduced
Fallen log	jump	brown rounded rect	0 m
Low branch	slide	green bar at head height	300 m
Spirit guardian (tall)	jump only	glowing wisp column	800 m
Spirit guardian (low)	slide only	hovering wisp disc	800 m
Rune trap	time it	floor spikes pulsing: 1.2 s cycle — 0.7 s dark (safe), 0.5 s lit (lethal)	1200 m

- Speed ramp as 05 (320→900 px/s). **Hearts: 2**; hit = −1 heart + 1.2 s invulnerability (blink); 0 → game over (deliberately softer than 05 — longer-form run).
- Palette shifts by depth tier (dawn → dusk → night forest).

Phase B — orbs & powers

- Orbs spawn in arcs/lines riding the spawner (arc of 5–7 over a jump, line of 4 on flat). Meter: 10 orbs → full. Power = next in a **fixed cycle** (readable): Double jump → Dash → Slow time (durations in the table below).

Power	Duration	Effect
Double jump	8 s	second mid-air jump
Dash	1.5 s	invulnerable, destroys the next obstacle hit (exactly one)

Power	Duration	Effect
Slow time	4 s	world $\times 0.6$ (audio pitch drops too)

- Power button (bottom-right, thumb reach) lights when meter full; keyboard E; optional auto-trigger setting.

Phase C — rune gates & shadow realm (the twist)

- Every **600 m $\pm$ 100** the path splits into two gates; a hint banner shows 2 s before. `gates.js` (pure) defines **5 rules**:

Rule	Hint text	Correct gate
Parity	"Even spirits walk the left path."	side matching <code>orbsSinceLastGate % 2</code>
Shape echo	"Follow the doubled symbol."	gate whose rune == the previous gate's rune
Negation	"The moon door lies."	NOT the gate matching the banner's shape hint
Sequence	"▲ ▲ ▲ ..."	gate continuing the shown 3-symbol pattern
Color trail	"Trust the last light you caught."	side matching last orb's color

- API: `makeGate(ruleId, rng, runState)  $\rightarrow$  { correctSide, hintText, runes }`; `resolveChoice(gate, side)  $\rightarrow$  'correct' | 'shadow'`. Rules cycle in order per run (learnable), order shuffles per run via seed.
- **Correct** $\rightarrow$ +100 points, continue forest. **Wrong** $\rightarrow$ **shadow realm**: 45 s, darker palette, obstacle density $\times 1.5$, orbs worth $\times 2$, guaranteed **+1 spirit shard** on survival; timer HUD countdown; auto-returns to forest (no soft-lock).

Phase D — characters & meta

- **Spirit shards**: `floor(meters/1000) + shadowSurvives + floor(correctGates/3)` per run.
- **Characters** (pick on menu; modifiers):

Character	Unlock	Modifier
Forest Spirit	start	power durations +50 %
Hunter	5 shards	every run starts with Dash charged
Monk	12 shards	Slow time also $\times 0.7$ obstacle spawn rate

- Save persists shards + unlocks forever (best scores separate).

3. Screens & UI

State	Elements
<code>menu</code>	Title, character cards (locked show shard cost), best distance/shards,  Run, mute
<code>playing</code>	World + HUD: hearts, orbs $\rightarrow$ power meter + button, distance, depth tier
<code>gate</code>	In-world split + hint banner (run continues — no separate screen)

State	Elements
shadow	Palette swap + "SHADOW REALM" banner + shard countdown
paused	Overlay (auto on <code>visibilitychange</code>)
gameover	Distance, orbs, gates correct, shards earned, unlocks, [Retry] [Character] [Share]

Virtual viewport 900×500 like 05; `?debug=1` draws hitboxes + gate rule id (dev aid).

4. Controls

- Jump: tap / Space / ↑ (double jump while empowered). Slide: swipe down / ↓ / S.
- Power: button (bottom-right) / E; auto-trigger accessibility setting.
- Swipe-up = jump, swipe-down = slide (pointer gesture thresholds: 30 px, 120 ms).

5. Data model (localStorage)

```
game:spirit-runner:save → {
  bestDistanceM: 2340, shards: 17, unlocked: ['spirit'],
  character: 'spirit', settings: { autoPower: false, muted: false }
}
```

6. File structure & function inventory

```
spirit-runner/
  index.html
  style.css
  engine.js  # forked from 05: loop, camera, parallax, input (jump+slide+swipes)
  core.js    # pure: obstacles, orbs, powers, gates resolution, shards – test surface
  gates.js   # pure: the 5 rune rules + hint text
  render.js  # palettes (forest/shadow), glow (lighter composite), particles, HUD
  main.js    # state machine + meta (characters, unlocks)
```

```
gates.js : RULES array, makeGate(ruleId, rng, runState), resolveChoice(gate, side),
hintFor(ruleId, runState). core.js : stepPlayer (jump/slide states), spawner
extensions ( spawnOrbArc, spawnTrap, spawnGuardian), applyPower(run, kind),
tickPowers(run, dt) (timers incl. character modifiers), shardsFor(run).
```

7. Edge cases

1. Gate while a power is active → power timers keep running through the split.
2. Wrong-gate during invulnerability → shadow realm still triggers (it's not damage).
3. Dash inside shadow realm destroys one obstacle only, as in forest.

4. Slow time stacks with Monk modifier → world $\times 0.6$, spawn $\times 0.7$ (multiplicative, capped once each — no double application).
5. Shadow realm timer always expires exactly (fixed-step accumulation; pause-safe); pays shards exactly once.
6. Unlocks persist across character switching; switching mid-menu only (never mid-run).
7. Rule sequence: 5 rules cycle; run seed shuffles the order; no rule repeats within a cycle; gates ≥ 600 m apart never overlap the shadow realm window (spawner defers).
8. Storage disabled → unlocks lost per session, playable.

8. Testing plan (gates.js + core.js)

- All 5 rules: `makeGate` + `resolveChoice` → correct side deterministic given a fixed `runState` (table tests per rule, 20 rng seeds each).
- Hint text present and never empty for any rule/state.
- Shard math: boundaries at 999/1000 m, 1/2/3 correct gates, shadow survive.
- Power timers: durations incl. Forest Spirit $\times 1.5$; Dash single-destroy assertion.
- Shadow realm: density multiplier applied; timer expiry; single shard payout.
- Spawner: gate deferral rule (§7.7) over 1,000 simulated runs.

9. Task breakdown

Phase A — core run (≈ 2 days)

- ☐ Fork engine from 05; slide + low-branch/log/guardian/trap obstacles; trap pulse cycle
- ☐ 2 hearts + invulnerability; depth palettes; pause/blur; menu/gameover; best distance

Phase B — orbs & powers (≈ 1.5 days)

- ☐ Orb arcs in spawner; power meter + cycle; DJ/Dash/Slow implementations
- ☐ Power button + auto-trigger setting; WebAudio (collect/power/hit) + mute

Phase C — rune gates & shadow realm (≈ 2 days)

- ☐ `gates.js` 5 rules + hints (+ tests); split-path rendering; resolution flow
- ☐ Shadow realm: palette, density $\times 1.5$, $\times 2$ orbs, shard payout, timed return, banner

Phase D — characters & meta (≈ 1.5 days)

- ☐ Shard earning; unlock thresholds; 3 character modifiers; picker UI
- ☐ Balance pass: median first-run 30–45 s, each gate rule learnable ≤ 3 exposures
- ☐ Share (`Ran {m} m as {character}... {pts}`); QA gate (master README §5)

P3 — polish

- ☐ Sprite/hand-drawn art slot (post-MVP); music track; daily depth seed; extra rules
- ☐ Achievements/leaderboards: only if isolation decision is reversed (master §7)

10. Acceptance criteria

- Each phase ships playable alone (A = good runner; A+B = powered; +C = full twist; D = meta).
- §8 tests green — all gate rules proven deterministic per state.
- Shadow realm never soft-locks; pays exactly once; always exits.
- 60 fps mid-range phone with particles + glow on; pause never desyncs timers.
- One-thumb: jump, slide, power reachable at 360 px width.

11. Deferred coupling (intentionally NOT built)

Footer/nav links, analytics events, platform achievements integration, leaderboards — per master README §7 (revisit only if the owner reverses isolation).

Source: plan/games markdown · generated 2026-09-09 · part of the 2D Games sample plans (plan/games/pdf).